The Appendix for Multi-branch of Attention Yields Accurate Results for Tabular Data

I. Appendices

A. Hyperparameter Tuning Space

Here we provide hyperparameter tuning spaces used for Optuna tuning for all methods in Section **??**.

1) MAYA: The hyperparameter tuning space for MAYA is presented in Table I.

2) AutoInt: The hyperparameter tuning space for AutoInt is presented in Table II.

3) TabTransformer: The hyperparameter tuning space for TabTransformer is presented in Table III.

4) SAINT: The hyperparameter tuning space for SAINT is presented in Table IV.

5) FT-Transformer: The hyperparameter tuning space for FT-Transformer is presented in Table V.

6) ExcelFormer: The hyperparameter tuning space for ExcelFormer is presented in Table VI.

7) AMFormer: The hyperparameter tuning space for AMFormer is presented in Table VII.

TABLE I
HYPERPARAMETER TUNING SPACE FOR MAYA.

	Parameter	Distribution
encoder	num_layers	UniformInt[1, 16]
	hidden_size	UniformInt[64, 256]
	num_heads	Categorical[1, 4, 8, 32]
	num_branch	UniformInt[2, 8]
	intermediate_factor	Categorical[0.8, 1, 1.3, 2]
	dropout	Uniform[0, 0.3]
	add_act	Categorical[true, false]
	if_bias	Categorical[true, false]
	act_type	Categorical[relu, prelu]
decoder	num_decoder_layers	UniformInt[1, 16]
	decoder_intermediate_factor	Categorical[0.8, 1, 1.3, 2]
	decoder_dropout	Uniform[0, 0.3]
	decoder_if_bias	Categorical[true, false]
	decoder_act_type	Categorical[relu, prelu]
	qk_shared_weights	Categorical[true, false]
others	batch_size	Categorical[1024, 2048, 4096]
	learning_rate	LogUniform[1e-5, 5e-3]
	weight_decay	Uniform[0, 0.3]

TABLE II
HYPERPARAMETER TUNING SPACE FOR AUTOINT.

Parameter	Distribution
n_layers	UniformInt[1, 6]
d_token	UniformInt[8, 64]
residual_dropout	{0, Uniform[0, 0.2]}
attention_dropout	{0, Uniform[0, 0.5]}
learning_rate	LogUniform[1e-5, 1e-3]
weight_decay	LogUniform[1e-6, 1e-3]

TABLE III
HYPERPARAMETER TUNING SPACE FOR TABTRANSFORMER.

Parameter	Distribution
dim	Categorical[32, 64, 128, 256]
depth	Categorical[1, 2, 3, 6, 12]
heads	Categorical[2, 4, 8]
attn_dropout	Uniform[0, 0.5]
ffn_dropout	Uniform[0, 0.5]
learning_rate	LogUniform[1e-6, 1e-3]
weight_decay	LogUniform[1e-6, 1e-1]

TABLE IV
HYPERPARAMETER TUNING SPACE FOR SAINT.

Parameter	Distribution
dim	Categorical[16, 32, 64]
depth	Categorical[4, 6]
heads	Categorical[4, 8]
attn_dropout	Uniform[0, 0.5]
ffn_dropout	Uniform[0, 0.5]
attn_type	{colrow, Categorical[colrow, row, col]}
learning_rate	LogUniform[3e-5, 1e-3]
weight_decay	{0, LogUniform[1e-6, 1e-4]}

TABLE V
HYPERPARAMETER TUNING SPACE FOR FT-TRANSFORMER.

Parameter	Distribution
n_layers	$\mathrm{UniformInt}[1, 4]$
d_token	$\mathrm{UniformInt}[64, 512]$
residual_dropout	$\{0, \mathrm{Uniform}[0, 0.2]\}$
attention_dropout	$\mathrm{Uniform}[0, 0.5]$
ffn_dropout	$\mathrm{Uniform}[0, 0.5]$
ffn_factor	$\mathrm{Uniform}[2/3, 8/3]$
learning_rate	$\mathrm{LogUniform}[1e\text{-}5, 1e\text{-}3]$
weight_decay	$\mathrm{LogUniform}[1e\text{-}6, 1e\text{-}3]$

TABLE VI
HYPERPARAMETER TUNING SPACE FOR EXCELFORMER.

Parameter	Distribution
n_layers	$\mathrm{UniformInt}[2, 5]$
d_token	$\mathrm{Categorical}[64, 128, 256]$
n_heads	$\mathrm{Categorical}[4, 8, 16, 32]$
residual_dropout	$\mathrm{Uniform}[0, 0.5]$
mix_type	$\mathrm{Categorical}[\mathrm{none}, \mathrm{feat_mix}, \mathrm{hidden_mix}]$
learning_rate	$\mathrm{LogUniform}[3e\text{-}5, 1e\text{-}3]$
weight_decay	$\{0, \mathrm{LogUniform}[1e\text{-}6, 1e\text{-}3]\}$

TABLE VII
HYPERPARAMETER TUNING SPACE FOR AMFORMER.

Parameter	Distribution
dim	$\mathrm{Categorical}[8, 16, 32, 64, 128]$
depth	$\mathrm{Categorical}[1, 4]$
attn_dropout	$\mathrm{Uniform}[0, 0.5]$
ffn_dropout	$\mathrm{Uniform}[0, 0.5]$
num_special_tokens	$\mathrm{UniformInt}[1, 4]$
learning_rate	$\mathrm{LogUniform}[1e\text{-}5, 1e\text{-}3]$
weight_decay	$\mathrm{LogUniform}[1e\text{-}6, 1e\text{-}3]$